

4 Division

Turn In

1. There were 30 ($150 \div 5$) dancers in each row.
2. With 15 ($150 \div 15$) dancers in each row, there would be 10 rows with 150 dancers.
3. If each row had nine dancers, there would be 16 rows and six dancers would be left.

Exercise 4A

2.

a.
$$\begin{array}{r} 1354 \\ 6 \overline{) 8125} \\ \underline{-6} \\ 21 \\ \underline{-18} \\ 32 \\ \underline{-30} \\ 25 \\ \underline{-24} \\ 1 \end{array}$$

Quotient = 1354
Remainder = 1

b.
$$\begin{array}{r} 257 \\ 9 \overline{) 2317} \\ \underline{-18} \\ 51 \\ \underline{-45} \\ 67 \\ \underline{-63} \\ 4 \end{array}$$

Quotient = 257
Remainder = 4

c.
$$\begin{array}{r} 1273 \\ 4 \overline{) 5092} \\ \underline{-4} \\ 10 \\ \underline{-8} \\ 29 \\ \underline{-28} \\ 12 \\ \underline{-12} \\ 0 \end{array}$$

Quotient = 1273
Remainder = 0

d.
$$\begin{array}{r} 1886 \\ 5 \overline{) 9432} \\ \underline{-5} \\ 44 \\ \underline{-40} \\ 43 \\ \underline{-40} \\ 32 \\ \underline{-30} \\ 2 \end{array}$$

Quotient = 1886
Remainder = 2

e.
$$\begin{array}{r} 1824 \\ 2 \overline{) 3648} \\ \underline{-2} \\ 16 \\ \underline{-16} \\ 04 \\ \underline{-4} \\ 08 \\ \underline{-8} \\ 0 \end{array}$$

Quotient = 1824
Remainder = 0

f.
$$\begin{array}{r} 151 \\ 7 \overline{) 1059} \\ \underline{-7} \\ 35 \\ \underline{-35} \\ 09 \\ \underline{-7} \\ 2 \end{array}$$

Quotient = 151
Remainder = 2

g.
$$\begin{array}{r} 505 \\ 8 \overline{) 4040} \\ \underline{-40} \\ 040 \\ \underline{-40} \\ 0 \end{array}$$

Quotient = 505
Remainder = 0

h.
$$\begin{array}{r} 1794 \\ 3 \overline{) 5383} \\ \underline{-3} \\ 23 \\ \underline{-21} \\ 28 \\ \underline{-27} \\ 13 \\ \underline{-12} \\ 1 \end{array}$$

Quotient = 1794
Remainder = 1

Application Based Questions

3.

Quantity of wheat in each bag = $1071 \text{ kg} \div 9$
= 119 kg

$$\begin{array}{r} 119 \\ 9 \overline{) 1071} \\ \underline{-9} \\ 17 \\ \underline{-9} \\ 81 \\ \underline{-81} \\ 0 \end{array}$$

4.

Number of storybooks = 1950

Number of cupboards = 8

Number of books placed in each cupboards = $1950 \div 8 = 243$

So, 243 storybooks can be placed in each cupboards and 6 storybooks will be left.

$$\begin{array}{r} 243 \\ 8 \overline{) 1950} \\ \underline{-16} \\ 35 \\ \underline{-32} \\ 30 \\ \underline{-24} \\ 6 \end{array}$$

5.

Cost of six chairs = ₹ 5868

Cost of each chair = ₹ $5868 \div 6$

= ₹ 978

So, the cost of each chair is ₹ 978.

$$\begin{array}{r} 978 \\ 6 \overline{) 5868} \\ \underline{-54} \\ 46 \\ \underline{-42} \\ 48 \\ \underline{-48} \\ 0 \end{array}$$

6. a. Total amount paid for eight blankets: ₹ 4208

Cost of each blanket = ₹ $4208 \div 8 = ₹ 526$

Ans: Each blanket costs ₹526.

$$\begin{array}{r} 8 \overline{) 4208} 526 \\ \underline{-40} \\ 20 \\ \underline{-16} \\ 48 \\ \underline{-48} \\ 00 \end{array}$$

b. We can help poor people shivering in cold by donating warm clothing and blankets to them.

Let's Try! 1

1. $763 \div 10$

Q = 76, R = 3

2. $350 \div 10$

Q = 35, R = 0

3. $4037 \div 100$

Q = 40, R = 37

4. $5005 \div 1000$

Q = 5, R = 5

Exercise 4B

1. a.
$$\begin{array}{r} 18 \\ 26 \overline{) 478} \\ \underline{-26} \\ 218 \\ \underline{-208} \\ 10 \end{array}$$

Quotient = 18
Remainder = 10

b.
$$\begin{array}{r} 42 \\ 16 \overline{) 680} \\ \underline{-64} \\ 40 \\ \underline{-32} \\ 8 \end{array}$$

Quotient = 42
Remainder = 8

c.
$$\begin{array}{r} 35 \\ 22 \overline{) 775} \\ \underline{-66} \\ 115 \\ \underline{-110} \\ 5 \end{array}$$

Quotient = 35
Remainder = 5

d.
$$\begin{array}{r} 49 \\ 15 \overline{) 735} \\ \underline{-60} \\ 135 \\ \underline{-135} \\ 0 \end{array}$$

Quotient = 49
Remainder = 0

$$\begin{array}{r} 17 \\ 39 \overline{) 684} \\ \underline{-39} \\ 294 \\ \underline{-273} \\ 21 \end{array}$$

Quotient = 17
Remainder = 21

$$\begin{array}{r} 40 \\ 19 \overline{) 767} \\ \underline{-76} \\ 07 \end{array}$$

Quotient = 40
Remainder = 7

$$\begin{array}{r} 33 \\ 63 \overline{) 2125} \\ \underline{-189} \\ 235 \\ \underline{-189} \\ 46 \end{array}$$

Quotient = 3
Remainder = 46

$$\begin{array}{r} 201 \\ 29 \overline{) 5839} \\ \underline{-58} \\ 039 \\ \underline{-29} \\ 10 \end{array}$$

Quotient = 201
Remainder = 10

$$\begin{array}{r} 52 \\ 58 \overline{) 3060} \\ \underline{-290} \\ 160 \\ \underline{-116} \\ 44 \end{array}$$

Quotient = 52
Remainder = 44

$$\begin{array}{r} 176 \\ 24 \overline{) 4226} \\ \underline{-24} \\ 182 \\ \underline{-168} \\ 146 \\ \underline{-144} \\ 2 \end{array}$$

Quotient = 176
Remainder = 2

$$\begin{array}{r} 51 \\ 84 \overline{) 4294} \\ \underline{-420} \\ 94 \\ \underline{-84} \\ 10 \end{array}$$

Quotient = 51
Remainder = 10

$$\begin{array}{r} 110 \\ 65 \overline{) 7153} \\ \underline{-65} \\ 65 \\ \underline{-65} \\ 03 \\ \underline{-0} \\ 3 \end{array}$$

Quotient = 110
Remainder = 3

Application Based Questions

$$\begin{array}{r} 23 \\ 21 \overline{) 488} \\ \underline{-42} \\ 68 \\ \underline{-63} \\ 5 \end{array}$$

Quotient = 23
Remainder = 5

$$\begin{array}{r} 26 \\ 35 \overline{) 916} \\ \underline{-70} \\ 216 \\ \underline{-210} \\ 6 \end{array}$$

Quotient = 26
Remainder = 6

3.

Number of students in school = 650

Number of students assigned to each teacher = 25

$\therefore$ The number of teachers required = $650 \div 25$
= 26

$$\begin{array}{r} 26 \\ 25 \overline{) 650} \\ \underline{-50} \\ 150 \\ \underline{-150} \\ 0 \end{array}$$

Ans: 26 teachers would be required for 650 students.

4. Number of shirts for one box = 24
 Total number of shirts = 886
 Number of boxes required: $886 \div 24$

$$\begin{array}{r} 36 \\ 24 \overline{) 886} \\ \underline{-72} \\ 166 \\ \underline{-144} \\ 22 \end{array}$$

Ans: For 886 shirts 36 boxes would be required and 22 would be left over.

5.

- Distance travelled in one month = 1320 km
 Working days in the months = 22
 $\therefore$ Distance travelled in one day = $1320 \text{ km} \div 22$
 $= 60 \text{ km}$

$$\begin{array}{r} 60 \\ 22 \overline{) 1320} \\ \underline{-132} \\ 00 \\ \underline{-00} \\ 0 \end{array}$$

Ans: In one day Rashmi travels 60 km.

It is better to walk for short distances as walking exercises the body and saves fuel.

6.

- Number of candles made = 750
 Number of candles in each set = 12
 $\therefore$ Number of sets/packets required = $750 \div 12$
 $= 62$

$$\begin{array}{r} 62 \\ 12 \overline{) 750} \\ \underline{-72} \\ 30 \\ \underline{-24} \\ 6 \end{array}$$

Number of candles left = 6

Ans: 62 packets were made and 6 candles were left unpacked.

7.

- Number of tiles bought = 1950
 Each room requires = 72 tiles
 $\therefore$ Number of classrooms that can be tiled = $1950 \div 72$
 $= 27$

$$\begin{array}{r} 27 \\ 72 \overline{) 1950} \\ \underline{-144} \\ 510 \\ \underline{-504} \\ 6 \end{array}$$

Number of tiles left = 6

Ans: 27 rooms were tiled and 6 were left over.

8. Cost of 18 litres of milk: ₹ 1170
 Cost of one litre: ₹ $1170 \div 18$

$$\begin{array}{r} 65 \\ 18 \overline{) 1170} \\ \underline{-108} \\ 90 \\ \underline{-90} \\ 00 \end{array}$$

In ₹ 65, Ambika can buy one litre of milk

In ₹ 715, Ambika can buy

$$715 \div 65$$

$$\begin{array}{r} 65 \overline{) 715} (11 \\ - 65 \\ \hline 65 \\ - 65 \\ \hline 00 \end{array}$$

Ans: Ambika can buy 11 litres of milk with the money she has.

9. No. of trees planted: 198

No. of students: 18

Number of trees planted by each student: $198 \div 18$

$$\begin{array}{r} 18 \overline{) 198} (11 \\ - 18 \\ \hline 18 \\ - 18 \\ \hline 00 \end{array}$$

Ans: a. Each student planted 11 trees.

b. Environment awareness

c. Planting trees helps to reduce global warming and carbon dioxide levels.

Exercise 4C

2.

- Vikram purchased 84 LED lights for electrification of seven houses. How many lights were fitted in each house?
- Manoj supplies 96 litres of milk to 8 houses every week. How much milk did each house receive?
- Rahul distributed 250 toffees in his class of 50 students. How many toffees did each student get?
- Anahita enrolled for a 121-hour swimming course spread over 11 months. How many hours of swimming class will she have to attend in one month?

Revision Exercises

Start Up

1.

- The quotient of a number divided by itself is one.
- When a number is divided by another number, the left over is known as remainder.
- It is not possible to divide any number by zero.
- Division is repeated subtraction.

2. a.
$$\begin{array}{r} 47 \\ 16 \overline{) 765} \\ \underline{-64} \\ 125 \\ \underline{-112} \\ 13 \end{array}$$

Quotient = 47
Remainder = 13

b.
$$\begin{array}{r} 40 \\ 21 \overline{) 842} \\ \underline{-84} \\ 2 \end{array}$$

Quotient = 40
Remainder = 2

c.
$$\begin{array}{r} 33 \\ 18 \overline{) 605} \\ \underline{-54} \\ 65 \\ \underline{-54} \\ 11 \end{array}$$

Quotient = 33
Remainder = 11

d.
$$\begin{array}{r} 21 \\ 43 \overline{) 925} \\ \underline{-86} \\ 65 \\ \underline{-43} \\ 22 \end{array}$$

Quotient = 21
Remainder = 22

e.
$$\begin{array}{r} 127 \\ 26 \overline{) 3309} \\ \underline{-26} \\ 70 \\ \underline{-52} \\ 189 \\ \underline{-182} \\ 7 \end{array}$$

Quotient = 127
Remainder = 7

f.
$$\begin{array}{r} 118 \\ 19 \overline{) 2255} \\ \underline{-19} \\ 35 \\ \underline{-19} \\ 165 \\ \underline{-152} \\ 13 \end{array}$$

Quotient = 118
Remainder = 13

g.
$$\begin{array}{r} 170 \\ 33 \overline{) 5612} \\ \underline{-33} \\ 231 \\ \underline{-231} \\ 02 \\ \underline{-0} \\ 2 \end{array}$$

Quotient = 170
Remainder = 2

h.
$$\begin{array}{r} 141 \\ 61 \overline{) 8607} \\ \underline{-61} \\ 250 \\ \underline{-244} \\ 67 \\ \underline{-61} \\ 6 \end{array}$$

Quotient = 141
Remainder = 6

3. a. $36 \times 20 = 720$

b. $8 \times 80 = 640$

c. $15 \times 7 = 105$

d. $33 \times 30 = 990$

Level Up

1. a. $550 \div 5$ gives 55 as quotient and 0 as remainder: False
 b. $6250 \div 5$ gives 125 as the quotient and 0 as the remainder: False
 c. $8400 \div 84$ gives 100 as quotient and 0 as the remainder: True

2. a. $548 \div 38$

$$\begin{array}{r} 14 \\ 38 \overline{) 548} \\ \underline{-38} \\ 168 \\ \underline{-152} \\ 16 \end{array}$$

Quotient = 14
Remainder = 16

Check: divisor $\times$ quotient + remainder = $38 \times 14 + 10$
 $= 380 + 152 + 16 = 548 = \text{dividend}$
 Thus, the answer is correct.

b. $667 \div 52$

$$\begin{array}{r} 12 \\ 52 \overline{) 667} \\ \underline{-52} \\ 147 \\ \underline{-104} \\ 43 \end{array}$$

Quotient = 12
Remainder = 43

Check: Divisor $\times$ quotient + remainder = $12 \times 52 + 43$
 $= 624 + 43 = 667 = \text{Dividend}$
 Therefore, answer is correct.

c. $2599 \div 45$

$$\begin{array}{r} 57 \\ 45 \overline{) 2599} \\ \underline{-225} \\ 349 \\ \underline{-315} \\ 34 \end{array}$$

Check: Quotient $\times$ divisor + remainder = $57 \times 45 + 34$
 $= 2565 + 34 = 2599 = \text{dividend}$
 Therefore, answer is correct.

Quotient = 57
 Remainder = 34

d. $1894 \div 76$

$$\begin{array}{r} 24 \\ 76 \overline{) 1894} \\ \underline{-152} \\ 374 \\ \underline{-304} \\ 70 \end{array}$$

Check: Quotient $\times$ divisor + remainder = $24 \times 76 + 70$
 $= 1,824 + 70 = 1894 = \text{dividend}$
 Therefore, answer is correct.

Quotient = 24
 Remainder = 70

3. a. Total number of coins = 172
 Number of coins in 1 page = 12
 Number of pages required to hold coins = $172 \div 12$

$$\begin{array}{r} 12 \overline{) 172} (14 \\ \underline{-12} \\ 52 \\ \underline{-48} \\ 4 \end{array}$$

Ans: In order to hold 172 coins, 14 pages would be full and 4 coins would remain.

- b. Total number of marbles = 165
 Number of friends = 13
 Number of marbles with each friend = $165 \div 13$

$$\begin{array}{r} 13 \overline{) 165} (12 \\ \underline{-13} \\ 35 \\ \underline{-26} \\ 9 \end{array}$$

Ans: Each friend got 12 marbles and 9 were left over.

- c. Number of players in 1 team = 11
 Total number of girls = 154
 Total number of teams = $154 \div 11$

Ans: With 154 girls 14 girls' cricket teams can be made.

$$\begin{array}{r} 11 \overline{) 154} (14 \\ \underline{-11} \\ 44 \\ \underline{-44} \\ 00 \end{array}$$

- d. Number of needy and deserving students: 12
 Total amount to buy stationery : ₹ 9612
 Amount each student gets = $9612 \div 12$

Ans : Each student will get ₹ 801.

$$\begin{array}{r} 12 \overline{) 9612} (801 \\ \underline{-96} \\ 001 \\ \underline{-0} \\ 12 \\ \underline{-12} \\ 00 \end{array}$$

Hots Questions

Number of big tanks = 25
 Small tank = 1
 Number of Gold fish = 456
 All fishes are divided equally in 25 tanks.

Gold fish : 456

$$\begin{array}{r} \textcircled{18} \leftarrow \text{Big tank} \\ 25 \overline{) 456} \\ \underline{-25} \\ 206 \\ \underline{-200} \\ \textcircled{6} \leftarrow \text{Small tank} \end{array}$$

Silver molly fish : 280

$$\begin{array}{r} \textcircled{11} \leftarrow \text{Big tank} \\ 25 \overline{) 280} \\ \underline{-25} \\ 30 \\ \underline{-25} \\ \textcircled{5} \leftarrow \text{Small tank} \end{array}$$

Glow light fish : 328

$$\begin{array}{r} \textcircled{13} \leftarrow \text{Big tank} \\ 25 \overline{) 328} \\ \underline{-25} \\ 78 \\ \underline{-75} \\ \textcircled{3} \leftarrow \text{Small tank} \end{array}$$

1. Total number of fishes in each big tank = $18 + 11 + 13 = 42$
2. Number of fishes in the small tank = $6 + 5 + 3 = 14$

Everyday Maths

1.

- a. Number of children taken to the amusement park = 595

Number of students in one bus = 35

Number of buses required = $595 \div 35 = 17$

Ans: 17 buses would be required to go to the amusement park.

- b. Amount collected from children below 15 years = ₹ 8500

Number of children who visited the part = 50

Ticket charges for each child = $8500 \div 50$

Ans: Ticket charges for each child were ₹ 170.